# 1901.2a

- IEEE 1901.2a 2013 is a wired technology that updates the original IEEE 1901.2 standard. This is the Narrow-band Power Line Communication standard (NB-PLC).
- Both continuous and direct current electric power lines are specified in IEEE 1901.2a.
- The data rate may be increased to 500 kbps.

- **IEEE 1901.2, a standard for low-frequency (below 500 kHz) narrowband power line communications (PLC). This technology uses existing electrical wiring to transmit data, supporting applications like smart grids, smart lighting, and industrial automation.**

- **NB-PLC is frequently used in the following scenarios:**
- **Smart metering –**
  The NB-PLC can automate the reading of utility meters such as electricity meters, gas meters and water meters.
- **Distribution automation –**
  The NB-PLC may be used for distribution automation, which helps monitoring and controlling all the power grid's components.
- **Public lighting –**
  It is a frequent application for NB-PLC, which includes lights seen in cities and along roadways, highways, and public spaces like parks.
- **Electric vehicle charging stations –**
  NB-PLC may be used to power charging stations for electric vehicles.
- **Micro-grids –**
  NB-PLC may be used to create micro grids, which are small energy grids that can run independently of the main grid.
- **Renewable energy –**
  The NB-PLC may be utilized in solar, wind, hydropower, and geothermal heat applications. There are several PLC standards, but the lack of a low-frequency PLC solution prompted the creation of IEEE 1901.2a. Below 500 kHz.

# Features

**Standardization and Alliances**

- Poor dependability, limited throughput, lack of management, and poor compatibility affected the earliest generations of NB-PLC systems.
- Orthogonal frequency-division multiplexing is used in the current NB-PLC specifications (OFDM). Digital data is encoded using OFDM on multiple carrier frequencies.

**Physical Layer**

- NB-PLC is defined for frequency bands from 3 kHz to 500 kHz.
- IEEE 1901.2a supports the largest set of coding and enables both robustness ~~and throughput.~~
- Tone maps and modulations for all bands, such as robust modulation (ROBO), differential binary phase-shift keying (DBPSK), differential quadrature phase-shift keying (DQPSK), differential 8-point phase shift keying (D8PSK), and optionally 16 quadrature amplitude modulation (16QAM) for some bands, are included in the standard.
- **Robust (Modulation) refers to a communication modulation technique, such as Gaussian Frequency Shift Keying (GFSK) modulation, which exhibits a high degree of resilience to interferences and noise in harsh industrial environments.**

# MAC Layer

- The IEEE 1901.2a MAC frame format is related to the IEEE 802.15.4 MAC frame.
- Information elements are one of the important components that have been moved over from IEEE 802.15.4e to IEEE 1901.2a.
- Segment Control is a field in IEEE 1901.2. This deals with the segmentation or fragmentation of upper-layer packets that are bigger than the MAC protocol data unit (MPDU) can carry.

| 3 | 2 | 1 | 0/2 | 0/2/8 | 0/2 | 0/2/8 | 0/5/6 10/4 | variable | variable | 2(By |
|---|---|---|---|---|---|---|---|---|---|---|
| Segment control | Frame control | Seq. number | Destination PAN Identifier | Destination. Address | Source PAN Identifier | Source Address | Auxiliary Security Header | Information Elements | Frame Payload | Frame check sequence |

| | | |
|---|---|---|
| MAC Header(MHR) | MAC Protocol Data unit(MPDU) payload | MAC Footer (MFR) |

MAC frame format for IEEE1901.2a

- The Frame Control field defines attributes such as frame type, addressing modes, and other control flags.
- The MAC Header field is composed of the Frame Control, Sequence Number and the Addressing fields.
- The Sequence Number field indicates the sequence identifier for the frame.
- The Addressing field specifies the Source and Destination PAN Identifier fields as well as the Source and Destination Address fields.
- The MAC Payload field varies by individual frame type.
- The MAC Footer field is a frame check sequence (FCS). A FCS is a calculation based on the data in the frame that is used by the receiving side to confirm the integrity of the data in the frame.

# Topology for IEEE 1901.2a :

- IEEE 1901.2a scenarios and deployment topologies are based on physical power lines.
- Signal transmission is limited by variables such as noise, interference, distortion, and attenuation, just as it is with wireless technology. Because these variables grow more prominent as distance increases, most NB-PLC implementations have a **mesh topology**.
- Mesh networks provide the benefit of allowing devices to relay traffic from other devices, allowing greater distances to be separated.
- The IEEE 1901.2a standard allows any upper-layer protocol to be used. As a result, IPv6 6LoWPAN and RPL (routing protocol) IPv6 variants are supported. These protocols allow mesh networks to be created over PLC using network layer routing.

# Constrained Nodes and Networks

**A constrained network exhibits below characteristics:**

- Low bit-rate/throughput
- High packet loss and high variability of packet loss
- Highly asymmetric link characteristics
- Lack of advanced network services like multi-cast

**The possible reasons for the characteristics of constrained network are below:**

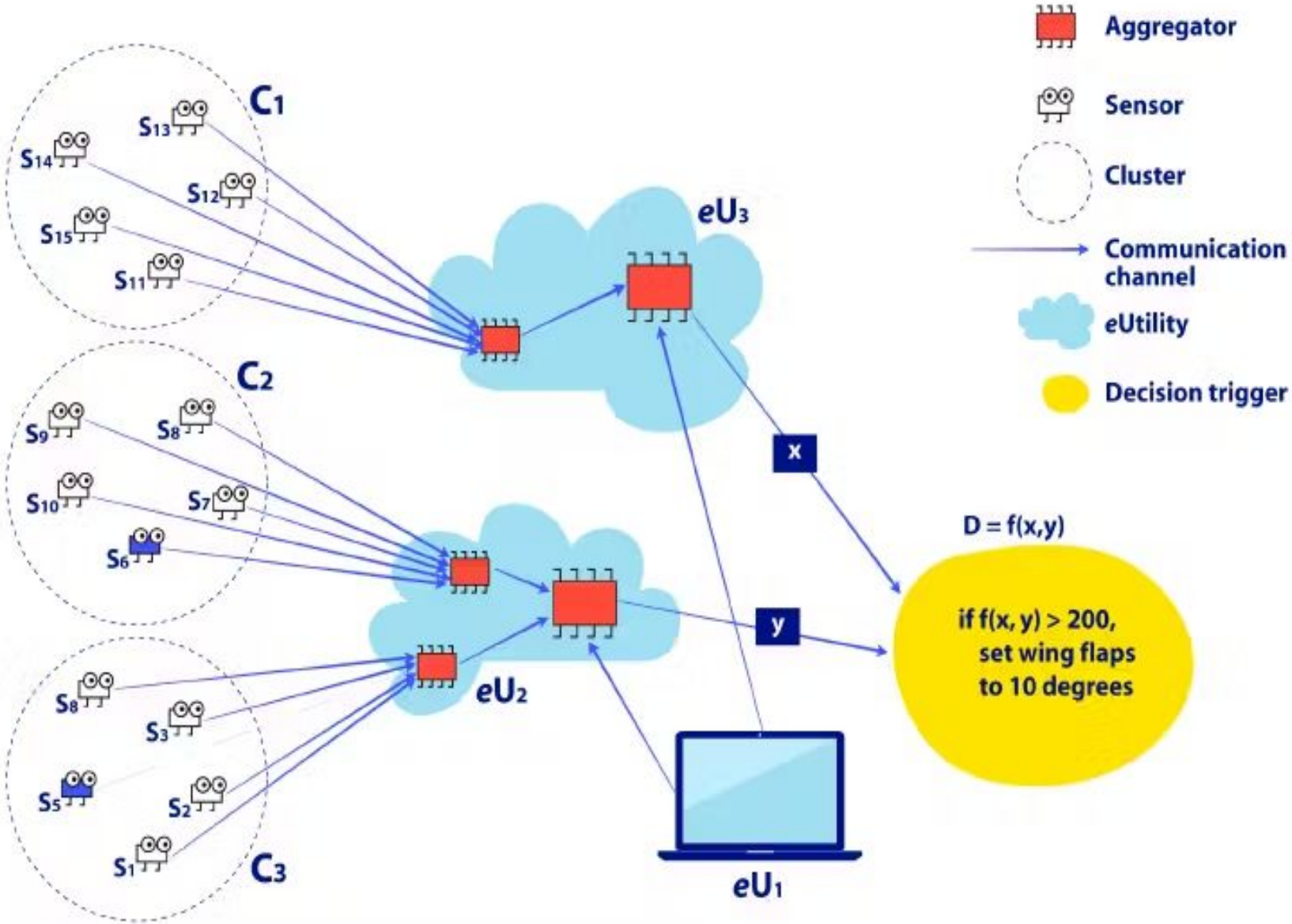
- Cost
- Constraints posed by the nodes
- Physical constraints, e.g., power constraints or environmental constraints
- i-Technology constraints such as using legacy technologies (outdated hardwares)

# Constrained node network

- A constrained network is composed of a significant portion of constrained nodes. Mostly, these **constrained node networks are deployed in the edge network of an IoT system.**

**Building Blocks of Constrained Node Network**

- **Sensors** – A sensor is an electronic utility that measures physical properties and gives electrical output. E.g. temperature sensor and acceleration sensor.
- **Actuators** - An actuator is a device which takes electronic input and gives physical output; e.g. motors.
- **Cluster** - A cluster is a group of sensors and actuators.
- **Communication Channel** - A communication channel is a medium through which data is transferred; e.g. wired or wireless.
- **Aggregators** – An aggregator is a device used to aggregate all the data from sensors and sometimes give the command to actuators. **It is the gateway device.**
- **eUtility** – It is a software or hardware or services which support aggregators for feeding data and computing.
- **Decision Trigger** – A decision trigger is a software which performs computing and takes action if needed.




Building blocks of constrained node network

# LORAWAN (Long Range Wide Area Network)

- LoRaWAN uses an unlicensed frequency band for long range communication.
- It can be used upto 3 miles and upto 10 miles in line of sight.
- Range can be extended in LoRaWAN using LoRaX.
- Since LoRaWAN devices operate at low power and low bandwidth the battery can run upto 10 years or more.

# Categories of LoRaWAN

**LORAWAN Class A devices**

- It has the longest battery life.
- It stays in sleep mode majority of the time
- It can receive messages only if it has sent messages.
- Example: fire alarm, flood detectors etc.

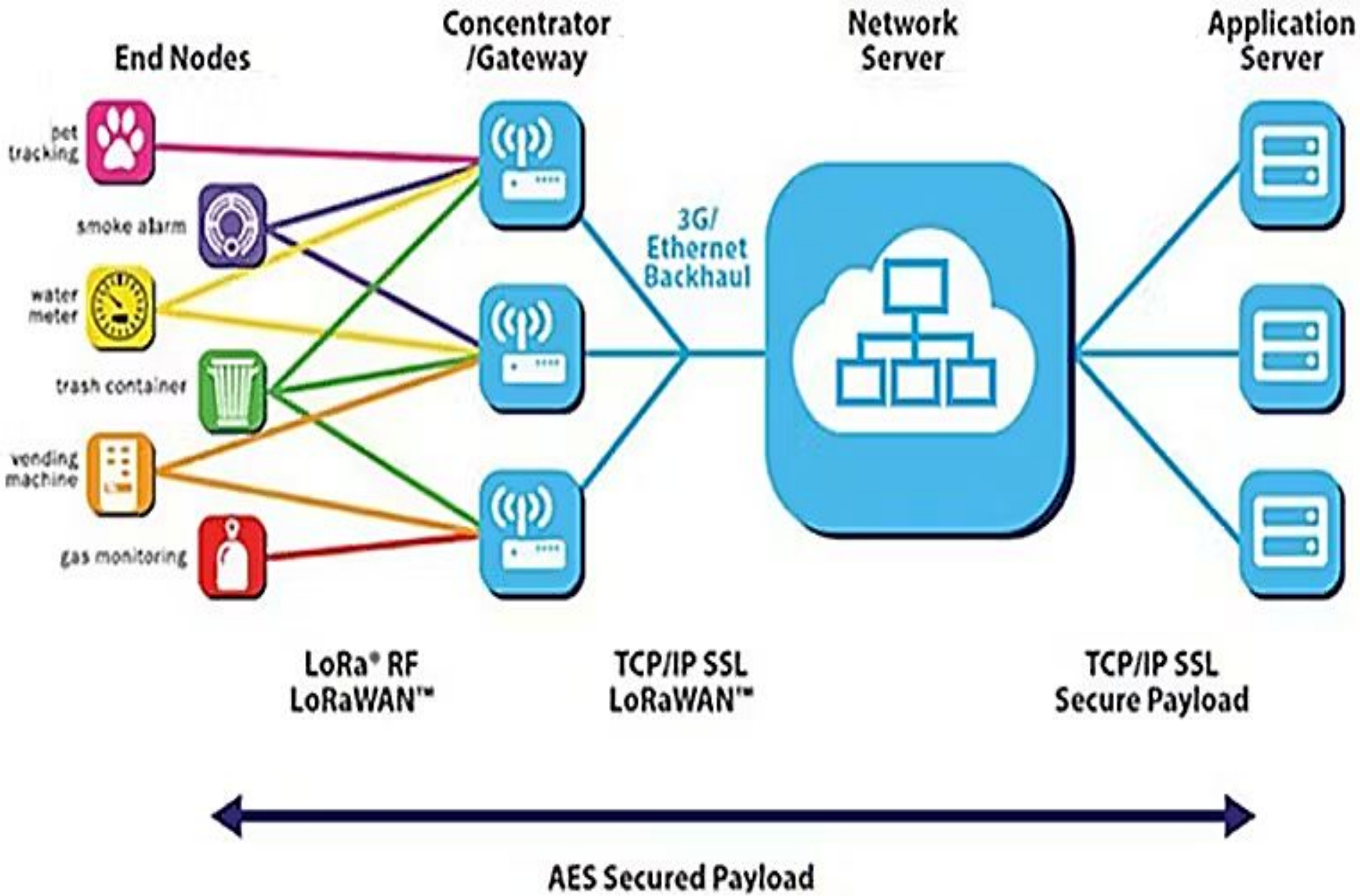
**LORAWAN Class B devices**

- It has the average battery life.
- It listens to the network periodically
- It can receive messages even if it has not sent messages.
- Example: metering temperature, humidity etc.

# LORAWAN Class C devices

- It has low battery life.
- It listens to the network continuously.
- It has minimum latency for messages.
- Example: traffic monitoring, real time applications etc.

# Network Server



- **The LoRa network has four elements:**
- **End-nodes** are elements of the LoRa ecosystem that gather sensor data and transmit/receive the data. They're generally remotely connected and are battery-powered.
- **The gateway** is a transparent bridge between the end-nodes and network server. Typically, end-nodes use LoRaWAN to connect to the gateway, while the gateway uses high-bandwidth networks such as Wi-Fi, Ethernet, or cellular to connect to the networks.
- **A network server** connects to multiple gateways. It gathers data from the gateways and filters out duplicate messages, decides which gateway should respond to end-node messages, and adapts data rates to extend battery life of end-nodes.
- **The application server** collects data from end-nodes and controls the actions of the end-node devices.

# Common applications of LoRaWAN

- Street Lighting
- Smart Parking
- Waste management
- Water level and flood management
- Smart Public Transit
- Air Quality Control

# Advantages

- High density network
- Low power consumption
- Higher data rates (~250kbps)
- Low cost
- Low bandwidth
- Extended coverage
- High mobility